

Koa Shen

B.S. Candidate in Mechanical Engineering · UC Santa Barbara

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October 1, 2026

Hiring Team

Neuralink

Fremont, CA

Re: Summer 2027 Medical Robotics Engineering Internship

Dear Neuralink Hiring Team,

I am a mechanical engineering student at UC Santa Barbara applying for a Summer 2027 opportunity in medical robotics. Neuralink's work interests me because it demands the kind of engineering I want to do: precise mechanisms and electromechanical systems designed around people, then tested carefully enough to earn trust in use. I would bring hands-on experience in motion systems, mechanical design, embedded controls, and fabrication, along with a strong desire to learn from a team working at the boundary of robotics and medicine.

At SLAC National Accelerator Laboratory, I built a Python digital-twin framework with live collision detection for tightly packaged, high-degree-of-freedom motion assemblies. I have used the model to guide a redesign of an X-ray optics assembly and performed structural checks for hardware protecting detectors valued at more than \$100,000. This work has taught me to think beyond whether a mechanism moves: clearances, failure modes, maintainability, and reliable documentation all matter when equipment must perform predictably.

In a personal robotics project, I designed and bench-tested a compact 15:1 cycloidal reducer and stepper package, then wrote embedded firmware using encoder feedback to detect missed steps and correct position error. With Gaucho Racing, I lead fabrication work, design driver-facing safety and ergonomics systems, and machine suspension components to bearing-interface tolerances within 0.0005 inches. Across these projects, I have learned to connect analysis and CAD to real hardware, listen closely to users, and iterate when testing reveals what the model missed.

I would welcome the opportunity to contribute that practical, detail-oriented approach to Neuralink's medical robotics work. I am especially eager to deepen my experience building precise, dependable electromechanical systems for applications where thoughtful engineering can make a meaningful difference to people's lives. Thank you for your consideration.

Sincerely,

Koa Shen